Azure IaaS, PaaS & SaaS – Easy Understanding Guide

A beginner-friendly reference for understanding Microsoft Azure cloud service models, with common examples and an AZ-900 memory trick.

1. What do IaaS, PaaS and SaaS mean?

Model	Simple meaning	What you mainly manage	Azure example
IaaS	Infrastructure as a Service – rent computing infrastructure.	VM, OS, applications, configuration and data.	Azure Virtual Machines
PaaS	Platform as a Service – Azure provides a managed platform to run apps/data.	Mostly your application and data.	Azure App Service / Azure SQL Database
SaaS	Software as a Service – ready-to-use software over the internet.	Mainly your users, settings and data.	Microsoft 365 / Microsoft Teams

2. IaaS – Infrastructure as a Service

Think: “I rent the infrastructure.”
Azure provides the basic infrastructure such as compute, networking and storage. You have more control, but you also have more things to manage.

Service	What it is
Azure Virtual Machines	Cloud-based virtual servers
VM Scale Sets	Manage groups of virtual machines
Managed Disks	Persistent disks for Azure VMs
Virtual Network (VNet)	Private networking in Azure
Load Balancer	Distributes traffic across resources
VPN Gateway	Connects networks securely using VPN
Application Gateway	Application-level traffic/load balancing
Azure Firewall	Managed network security firewall
ExpressRoute	Private connection between on-premises networks and Azure
Azure Bastion	Secure browser-based access to VMs

3. PaaS – Platform as a Service

Think: "I deploy my application; Azure manages much of the platform."
PaaS reduces infrastructure management so developers can concentrate on applications, databases, APIs, integration and data.

Service	What it is
Azure App Service	Managed platform for web apps and APIs
Azure Functions	Serverless, event-driven application code
Azure SQL Database	Managed relational database
Azure Cosmos DB	Managed distributed NoSQL database
Azure Database for PostgreSQL	Managed PostgreSQL database
Azure Container Apps	Managed platform for containerized applications
Azure Kubernetes Service (AKS)	Managed Kubernetes service
Azure Logic Apps	Workflow and integration service
Azure API Management	Publish, secure and manage APIs
Azure Service Bus	Messaging service for applications
Azure Event Grid	Event routing and event-driven integration
Azure Data Factory	Cloud data integration/ETL service
Azure Synapse Analytics	Analytics and data warehousing platform
Azure Machine Learning	Managed machine learning platform

4. SaaS – Software as a Service

Think: "I just use the software."
The provider manages the application, infrastructure, updates and most of the platform. You normally use the service through a web browser or application.

Service	Common use
Microsoft 365	Productivity suite: Word, Excel, PowerPoint, etc.
Microsoft Teams	Chat, meetings and collaboration
Microsoft Outlook	Email and calendar
OneDrive	Cloud file storage and sharing
SharePoint	Document management and collaboration
Power BI	Business intelligence and dashboards
Dynamics 365	CRM/ERP business applications
Microsoft Intune	Cloud-based endpoint/device management
Power Automate	Cloud workflow automation
Power Apps	Low-code application platform/service
Azure DevOps Services	Cloud development and DevOps services

5. Easy real-world analogy

IaaS = Renting an empty house: You get the building and utilities, but you manage much of what goes inside.
PaaS = Renting a furnished apartment: Much of the infrastructure is already managed; you focus on using the space for your purpose.
SaaS = Staying in a hotel: The provider manages almost everything; you simply use the finished service.

6. AZ-900 exam memory trick

IaaS → **VM** — You manage more.
PaaS → **App Service / Azure SQL** — Azure manages the platform.
SaaS → **Microsoft 365 / Teams** — Ready-to-use software.

7. Important note

Some Azure services do not fit perfectly into a single textbook category because cloud services can have different management responsibilities depending on how they are used. For AZ-900, focus on the core concept: **IaaS gives you the most infrastructure control, PaaS removes much of the infrastructure management, and SaaS gives you a ready-to-use application.**